

AI509: NLP, Fall 2026

Reasoning in Latent Space

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Keywords

Recurrent Depth / Latent Recursion

Looped Transformer

Hierarchical Reasoning Models

Soft Chain-of-Thought

Coconut

Full bandwidth transformer

System One / Decision Models

**Key Challenge: Reasoning
in token space is expensive.**

Recurrent Depth

(aka Latent Recursion)

Recurrent Depth

Key idea: Apply a transformer block repeatedly to shape the residual stream → Parameter sharing along the depth axis.

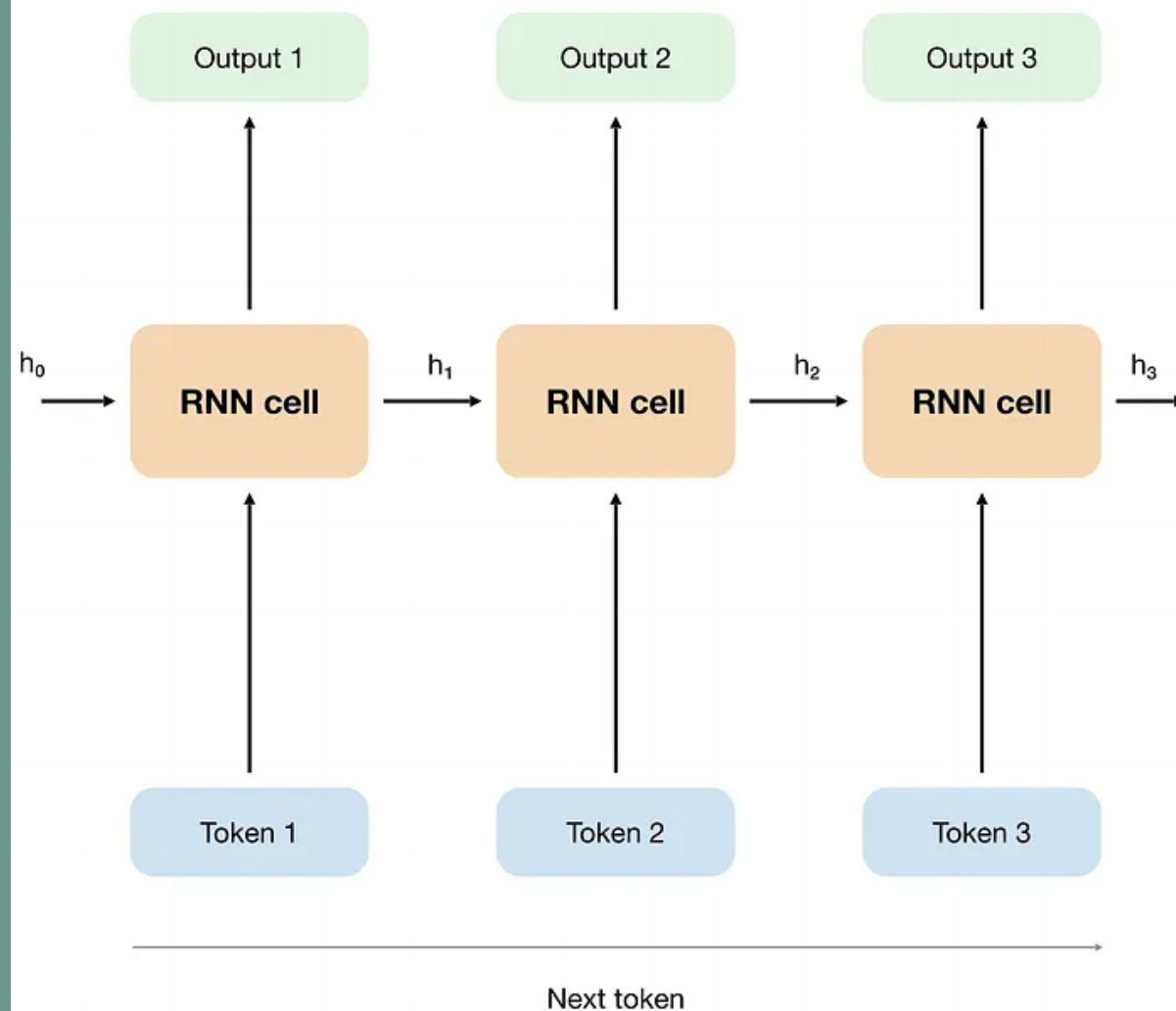
GPT-6 Astra is rumored to be a Recurrent Depth model¹

1. <https://magazine.sebastianraschka.com/p/gpt-6-astra-looped-transformers-and> ↩

Looped Transformers

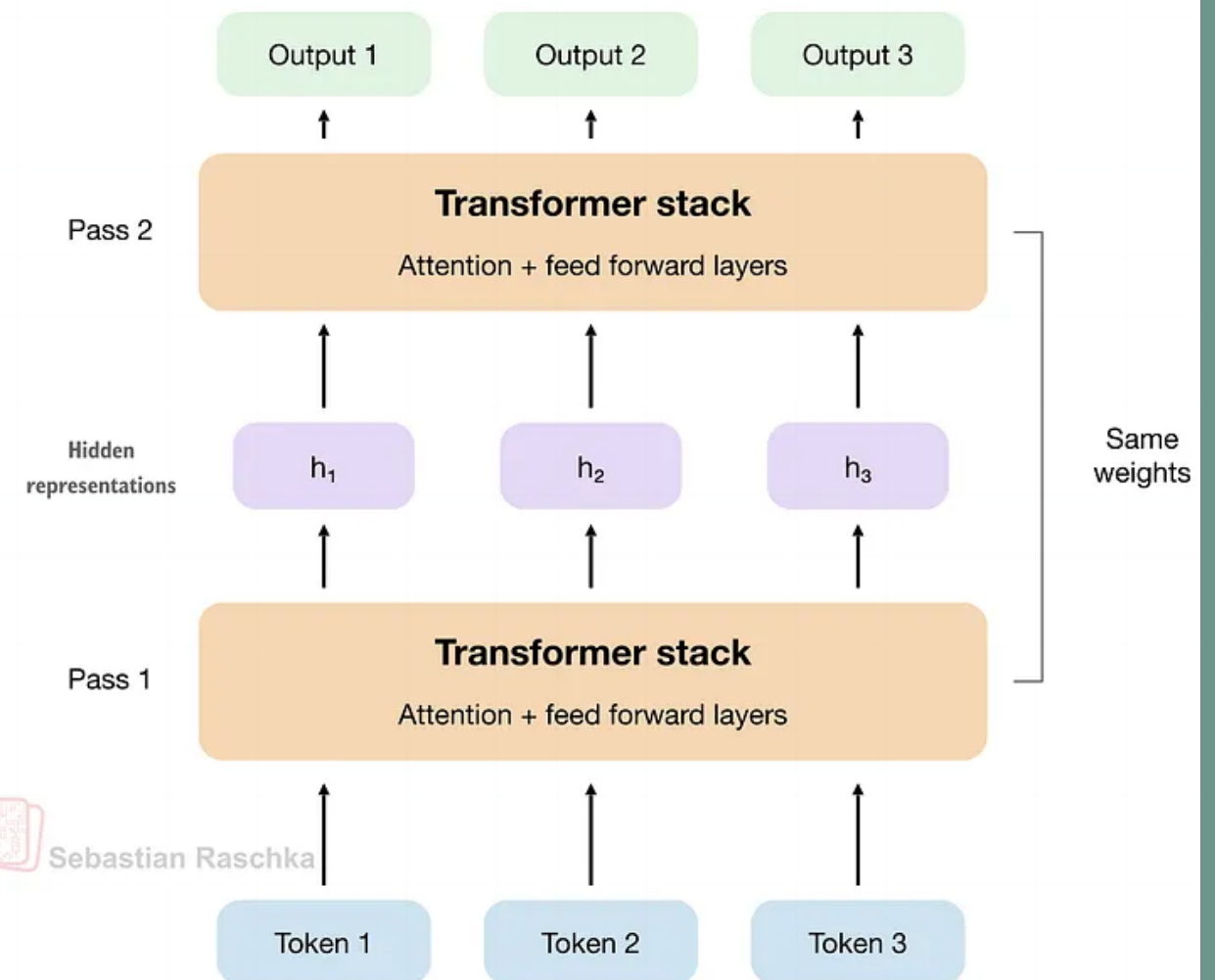
(a) RNN

Shared weights across time steps



(b) Looped transformer

Shared weights across depth



Sebastian Raschka

Notation for Transformers in a nutshell

Embedding $\text{embed}(\cdot) : \mathbb{R}^{|V|} \rightarrow \mathbb{R}^d$

Model $\mathcal{M}^L(\cdot) = \mathcal{T}_{\theta_L} \circ \mathcal{T}_{\theta_{L-1}} \circ \cdots \circ \mathcal{T}_{\theta_0}(\cdot)$

where each $\mathcal{T}_{\theta_i}$ is a transformer block.

LM head $\text{unembed}(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^{|V|}$

Standard Transformer: $\text{unembed} \circ \mathcal{M}^L \circ \text{embed}(\cdot)$

Looped Transformer

$$\text{unembed} \circ \underbrace{\mathcal{M}^L \circ \mathcal{M}^L \dots \circ \mathcal{M}^L}_{\text{Repeat } t \text{ times}} \circ \text{embed}(\cdot)$$

Looped Transformer with Prelude and Coda

$$\text{unembed} \circ \mathcal{C}^L \circ \underbrace{\mathcal{M}^L \circ \mathcal{M}^L \dots \circ \mathcal{M}^L}_{\text{Repeat } t \text{ times}} \circ \mathcal{P}^L \text{ embed}(\cdot)$$

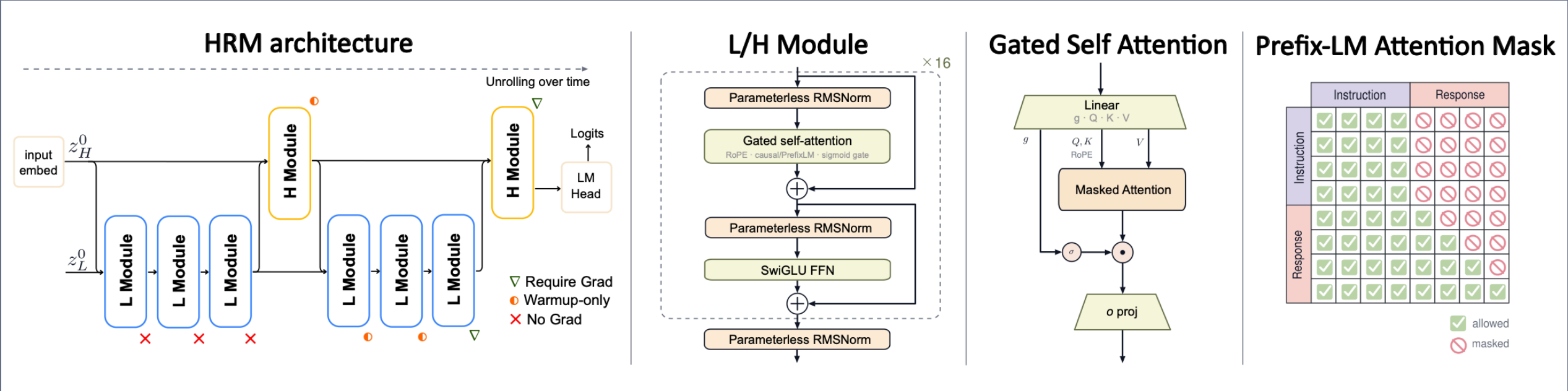
Adaptive computation (aka dynamic depth)

Idea: Some tasks require more reasoning than others.

Ingest a cross-entropy training objective after each recurrent step. Prepare the model to dynamically adjust depth.

Then train a linear module "early-exit" that decides when to exit the recurrence based on the current hidden state.

Hierarchical Reasoning Models (HRM)



Hierarchical Reasoning Model

Two modules $\mathcal{L}$ (fast, low-level reasoning)

and $\mathcal{H}$ (slow, higher-level reasoning)

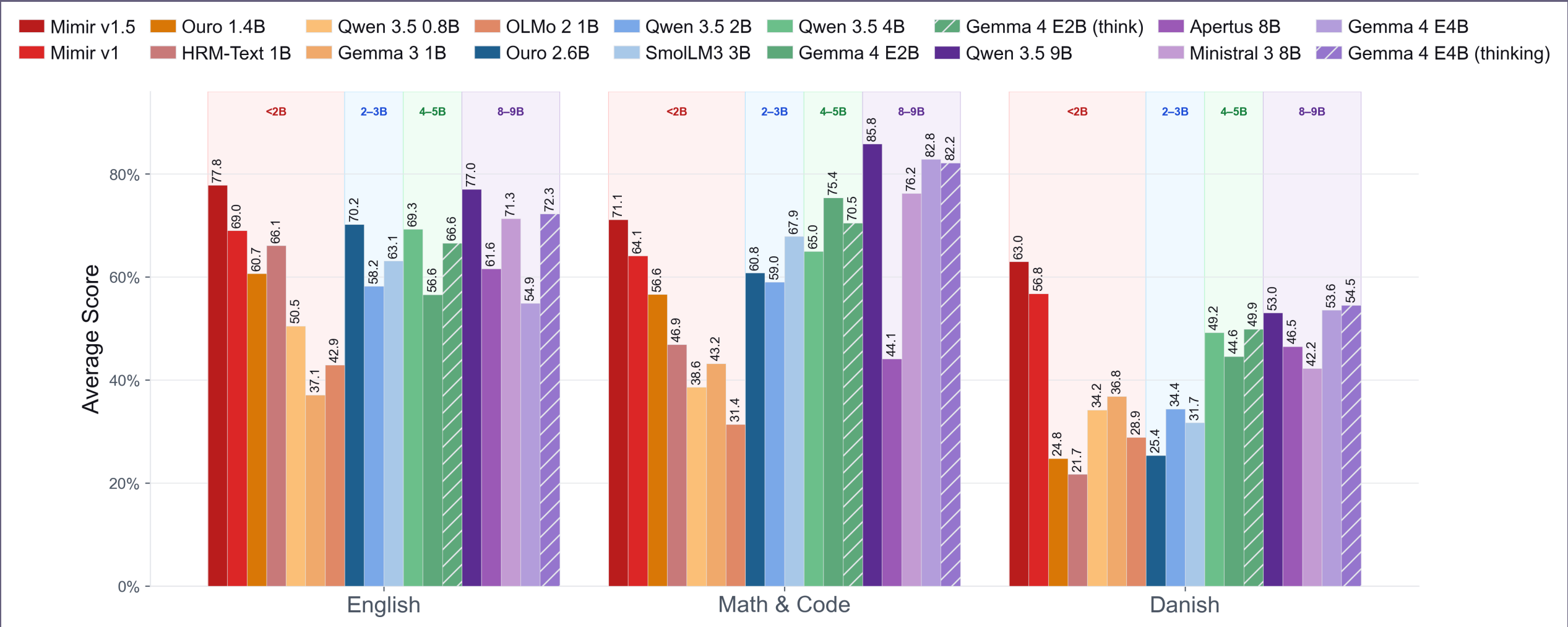
Interleave them via $\mathcal{L}\mathcal{L}\mathcal{L}\mathcal{H}\mathcal{L}\mathcal{L}\mathcal{L}\mathcal{H}$

$$\text{HRM}(\cdot) = \underbrace{\mathcal{H}^L \circ \mathcal{L}^L \circ \mathcal{L}^L \circ \mathcal{L}^L}_{\text{Repeat 2 times}}(\cdot)$$

Input to each module is $z = z_H + z_L$ where z_H is the latest available $\mathcal{H}$ state init. with $\text{embed}(\cdot)$, and z_L is the latest available $\mathcal{L}$ state.

Backpropagation only runs through the final HLLLH layer applications.

DFM Mimir v1.5



How did we do that?

Mainly data work. 161 synthetically generated datasets, turning raw data into **instruction data**.

Plus some tricks to stabilize the loss when gradually increasing backpropagation depth.

Further reading: [Sebastian Raschka on Looped Transformers](#)

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- ☐ Coconut 
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Soft chain-of-thought models

Another way to reason in latent space

Standard autoregressive transformer

Sample x_t from

$$\text{unembed} \circ \mathcal{M} \circ \text{embed}(x_{<t})$$

Then continue with predicting x_{t+1} by feeding back x_t to the input (**autoregressive** decoding)

Chain of Continuous Thought (Coconut 🥥)

Take the **latent state** z_t from

$$\mathcal{M}^L \circ \text{embed}(x_{<t}),$$

then continue with predicting z_{t+1} from

$$\mathcal{M}^L ([\text{embed}(x_{<t}); z_t])$$

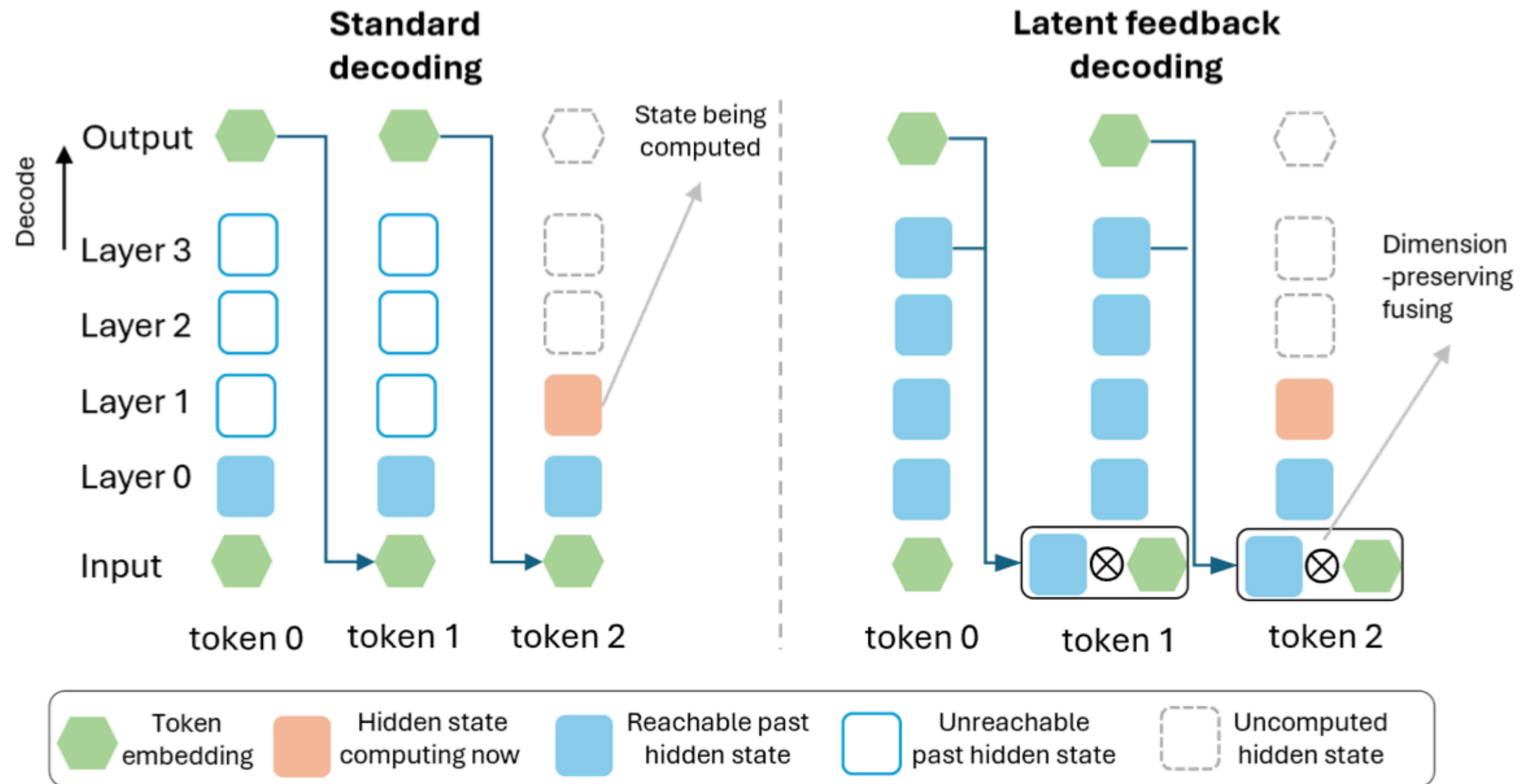
Note: No sampling. Switching between latent space and token space via special tokens.

Full bandwidth transformer

Coconut's switching between the soft chain of thought and the hard chain of thought.

Key idea: Use latent feedback together with the sampled token.

Full bandwidth transformer



Take the **latent state**

$$z_t = \mathcal{M}^L \circ ([\text{embed}(x_{<t}); z_{<t}]) ,$$

and sampled token $x_t \sim \text{unembed}(z_t)$

Continue with producing x_{t+1}, z_{t+1} from

$$\mathcal{M}^L ([\text{embed}(x_{\leq t}); z_{\leq t}])$$

Full bandwidth = Autoregression in **both** latent and token space.

Summary Reasoning in Latent Space

Recurrent depth: Apply the same transformer stack repeatedly.

Soft chain of thought: Carry over the final hidden state, instead of sampling a token during autoregression.

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Key Challenge: Reasoning in token space is expensive.

Observation: Many tasks do not need extensive reasoning.

→ **System One models (aka Decision Models)**

"System One" borrows from Kahneman's dual-process theory: fast, bounded, pattern-matching judgment, as opposed to slow deliberative reasoning ("System Two", i.e. chain-of-thought LLMs).

Key ingredients of System One Models

Fast, calibrated, typed decisions.

Calibrated: The decision comes with an estimated probability that reflects the accuracy of the model.

Fast: No chain of thought. A single forward pass should give the results.

- **Input:** unstructured text/state (documents, records, agent traces), plus one or more *questions*.
- **Output:** floats, not text. Three primitives (shared by Jev and Laya):
 - **Noul:** yes/no proposition, probability in $[0, 1]$ (Bernoulli).
 - **Choice:** pick from a finite option set, returns a distribution plus confidence.
 - **Score:** rating on an ordered rubric/numeric range, probability-weighted.
- **Typed:** By design the model output will never violate the expected schema.

How to train a System 1 model

- In principle, you can extract probabilities from the logics of a language model, even a BERT-style model.
- But extra training for calibration is important. There are many different ways to achieve this.
- Further reading: [razorback16/openjev](https://razorback16.github.io/openjev)

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